

# Marketing Analytics

## Chapter 3 · Uncertainty

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# Before we start

## Paper quiz on Chapter 2 (*10 min*)

- Individual, closed book
- Counts toward participation

## Assignment 1 is running

- Deadline: Group A Mon, Oct 26 · Group B Wed, Oct 28
- Debrief at the start of next session
- Questions? → forum on Canvas

# Today's session

## Part A — Uncertainty (*this deck*)

- 3.1 Is the difference real?
- 3.2 Populations and samples
- 3.3 Sampling variation
- 3.4 The standard error
- 3.5 The central limit theorem
- 3.6 Confidence intervals
- 3.7 Shares and the margin of error
- 3.8 Uncertainty in R

## Part B — Experiments I (*next deck*)

Comparing two groups: is the difference larger than sampling variation alone would produce?

**Driving question:** *How sure can I be?* Every number from a sample is an estimate — today we learn to say **how good** an estimate it is.

# 3.1 Is the difference real?

# An A/B test at a streaming service

- New feature **Smart Mix**: a personalized daily mix on the home screen
- **400 randomly selected users**, randomly split: 200 see Smart Mix, 200 the usual home screen
- Outcome: **listening hours** in the four weeks after the start

```
1 ab_test |>
2   group_by(group) |>
3   summarise(n = n(), mean = mean(hours), sd = sd(hours))
```

```
# A tibble: 2 × 4
  group      n mean  sd
<fct> <int> <dbl> <dbl>
1 Smart Mix   200  23.2  16.2
2 Control     200  20.0  13.7
```



# Would you roll it out?

```
1 t.test(hours ~ group, data = ab_test)
```

Welch Two Sample t-test

```
data:  hours by group
t = 2.1617, df = 386.96, p-value = 0.03125
alternative hypothesis: true difference in means between group Smart Mix and group Control is not equal to 0
95 percent confidence interval:
 0.293571 6.196096
sample estimates:
mean in group Smart Mix    mean in group Control
      23.21980             19.97497
```

Smart Mix users listened **3.2 hours more**. With **another 400 users**, would the difference be the same? Could Smart Mix do **nothing at all**? What do  $t$ ,  $p$ -value, and “95 percent confidence interval” tell us?

# Why statistical inference?

- The team observed **400 users** — the decision concerns **all users** (millions)
- 400 users = **sample**; all users = **population**
- The group means — and their difference — are **estimates**: another random sample, another number

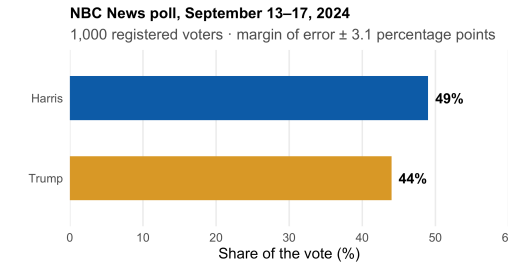
**Randomization** makes the comparison **fair** (§1.6).

**Statistical inference** tells us whether it is **precise enough** to tell a real effect from the noise of sampling.

## “Plus or minus 3.1 percentage points”

“The poll was conducted Sept. 13–17 and surveyed 1,000 registered voters nationally. The margin of error is plus or minus 3.1 percentage points.”

— NBC News, September 2024

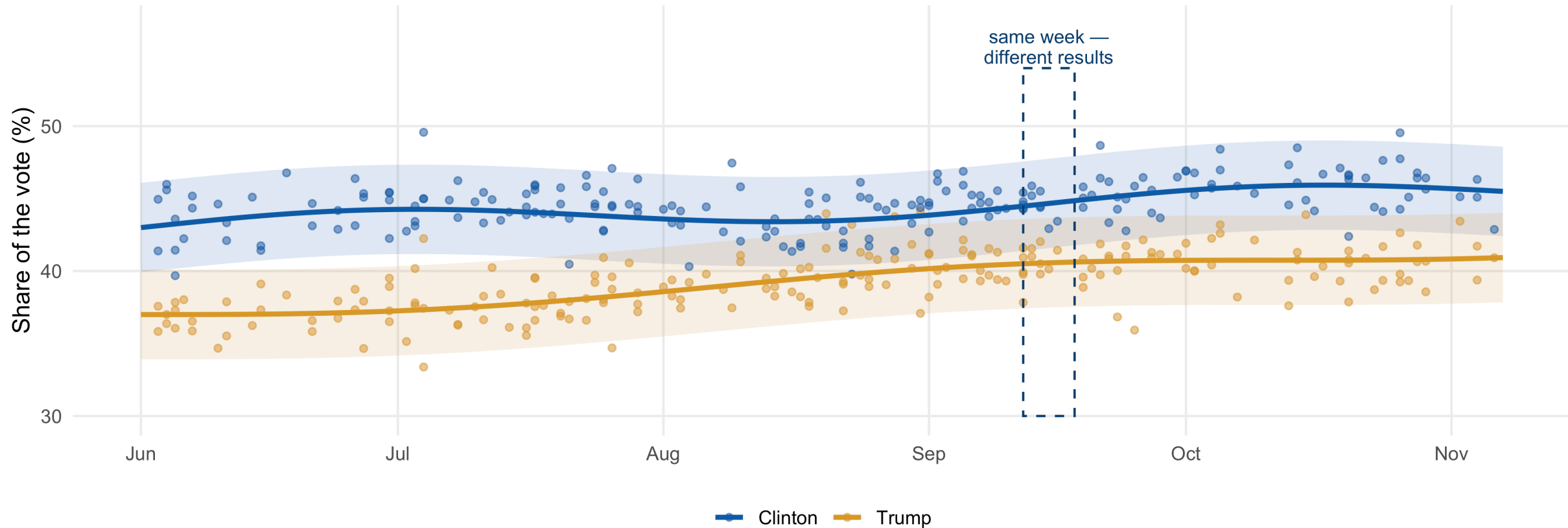


The same problem in politics: more than 150 million people voted. How can **1,000** respondents tell us anything? Where does  $\pm 3.1$  come from — and why not survey 10,000?

# Many polls, one truth

## Many polls, one truth (simulated)

Dots: individual polls of 600–1,500 voters · lines: true support · bands:  $\pm 3.1$  points around the truth



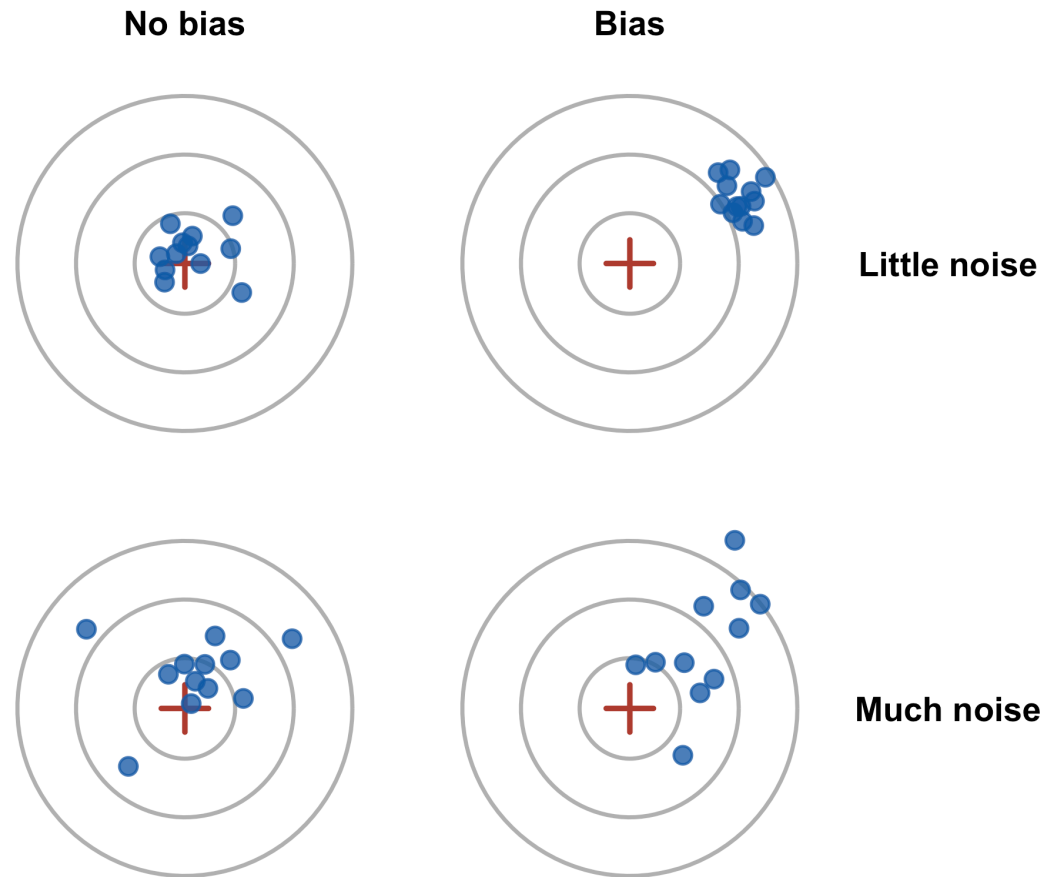
Polls taken in the **same week** come to **different results** — each one is a different random sample. Most of them stay within  $\pm 3$  **points** of the truth. That is what the margin of error describes.

# And then the polls were wrong

- 2016: state polls in several Midwestern states **underestimated Trump's support by more than their margins of error**
- Post-election analyses: late deciders; **voters without a college degree underrepresented** and not corrected for
- The margin of error did not warn — because it only covers **one kind** of error

Kennedy et al. (2018), "An Evaluation of the 2016 Election Polls in the United States", *Public Opinion Quarterly*, 82(1), 1–33.

# Two kinds of error



## Random error (sampling variation)

- Each sample is a slightly different mix of people
- **Quantifiable** — and **shrinks** with larger samples
- The margin of error is about this kind

## Systematic error (bias)

- The sample differs from the population *systematically*
- A larger sample **does not help** — “more confidently wrong” (§1.6)

*In the A/B test, randomization rules out bias in the comparison — random error remains.*

# Two examples, one question

## A/B test

- Outcome: listening hours → **continuous**
- We compare **means**

## Election poll

- Outcome: vote → **categorical**
- We look at **shares**

Same question in both cases: **how far can the number from our sample be from the true value in the population?** First for means (3.3–3.6), then for shares (3.7) — and then the A/B test output stops being a mystery.

## 3.2 Populations and samples

# From the sample to the population

- **Population:** the whole group we want to learn about (all voters, all customers, all WU students)
- **Census:** data from every member — rare, expensive, often impossible
- **Sample:** a subgroup we actually observe
- **Statistical inference:** generalizing from the sample to the population

|                    | Sample<br>(statistic) | Population<br>(parameter) |
|--------------------|-----------------------|---------------------------|
| Size               | $n$                   | $N$                       |
| Mean               | $\bar{x}$             | $\mu$                     |
| Standard deviation | $s$                   | $\sigma$                  |
| Share              | $\hat{p}$             | $p$                       |

Parameters: what we want to know. Statistics: what we can compute.

# How the sample is drawn matters

|             | Probability sampling                              | Non-probability sampling                    |
|-------------|---------------------------------------------------|---------------------------------------------|
| Principle   | Known chance of selection; <b>random</b>          | Convenience or researcher's judgment        |
| Techniques  | Simple random · stratified · cluster · systematic | Convenience · quota · judgmental · snowball |
| Uncertainty | Can be <b>quantified</b>                          | Cannot be quantified reliably               |

**Typical sources of selection bias:** *under-coverage* (not in the sampling frame) · *non-response* (non-respondents differ) · *voluntary response* (only the strongly opinionated answer)

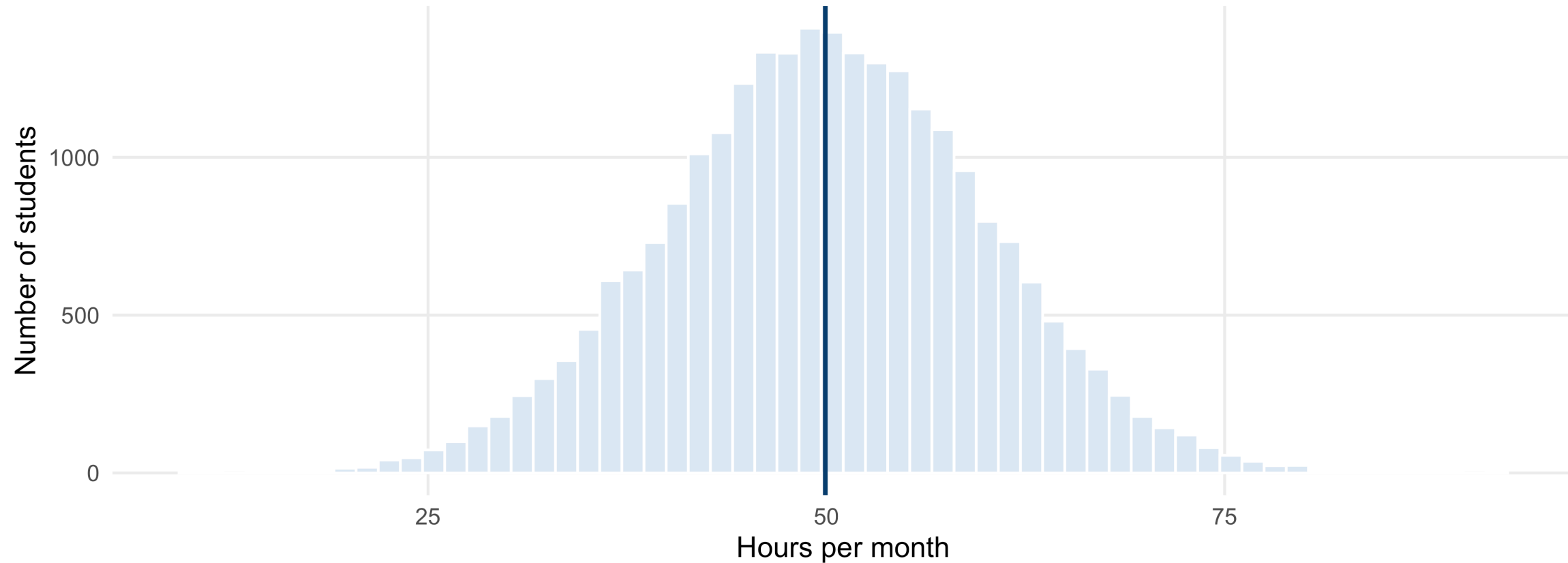
**Your group project** uses a convenience sample — fine for learning, but be honest about it. The good news: your **experiment** is still fair, because conditions are randomized *within* the sample (§1.6).

## 3.3 Sampling variation

# A world in which we know the truth

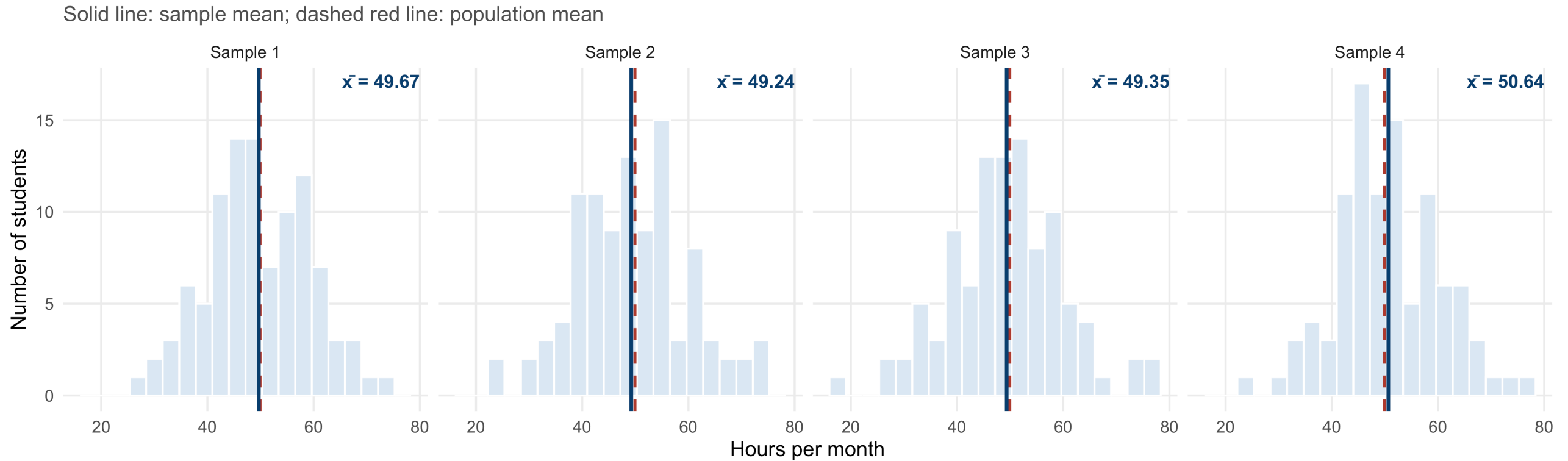
## Monthly listening time of all 25,000 WU students

Population mean  $\mu = 49.93$  hours; population SD  $\sigma = 10.02$  hours



Monthly listening time of all 25,000 WU students — simulated, so we know the true values. In reality we never do: that is why we sample.

# One sample, then another



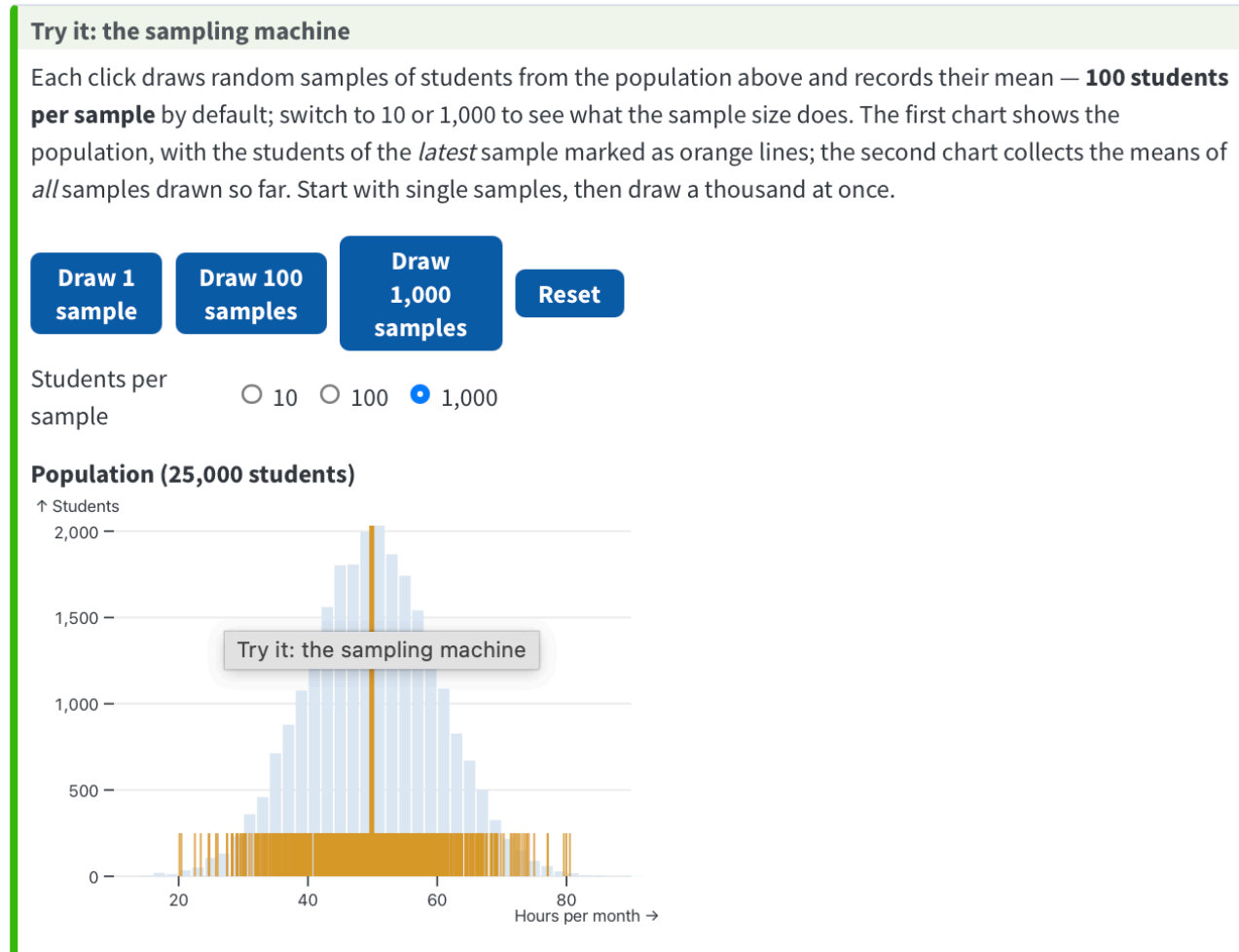
**Sampling variation:** each random sample of 100 students gives a slightly different mean. Not a mistake — the unavoidable price of looking at a part instead of the whole. The question is only: **how large** is it?

# Live demo: the sampling machine

Course book, §3.3 — draw 1, then 100, then 1,000 samples and watch the means pile up.

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-sampling>

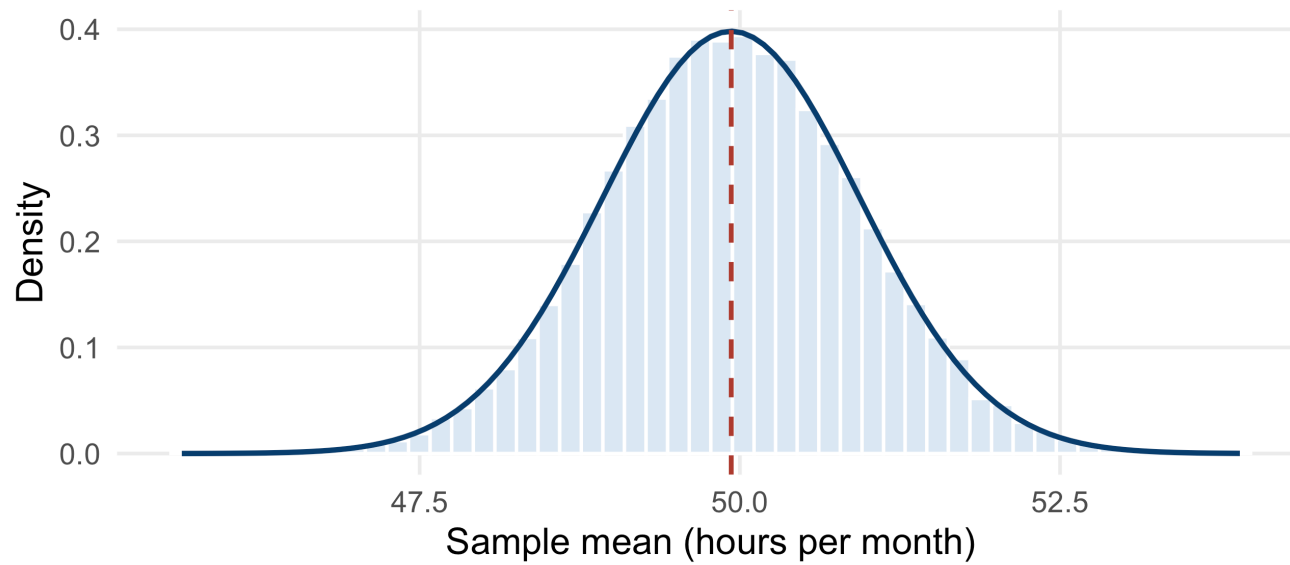
Before clicking “1,000 samples”: where will the sample means pile up — and how far from the true mean will the most extreme ones be?



# The sampling distribution

**Means of 20,000 samples with  $n = 100$  students each**

Mean of the sample means = 49.93; SD of the sample means = 1.00



The means of *many* samples:

1. **Centered on the truth** — no over- or underestimation on average
2. **A known width** — the *standard error*
3. **A known shape** — normal (even for non-normal data, \$3.5)

We only ever have **one** sample — but these three properties tell us how much to trust it.

## 3.4 The standard error

# How precise is a sample mean?

The width of the sampling distribution:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

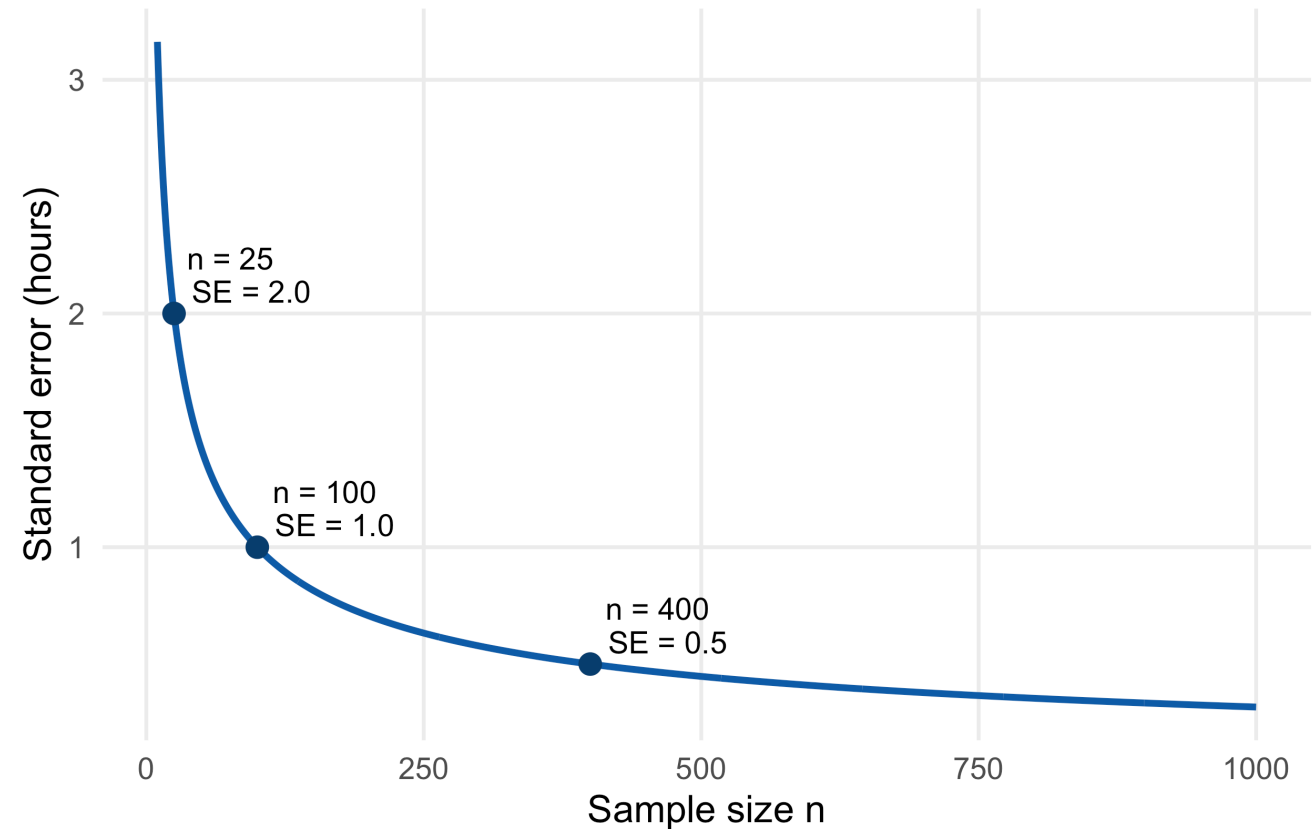
- $\sigma$ : variability in the population
- $n$ : sample size

Our example:  $10 / \sqrt{100} = 1.00$  hours

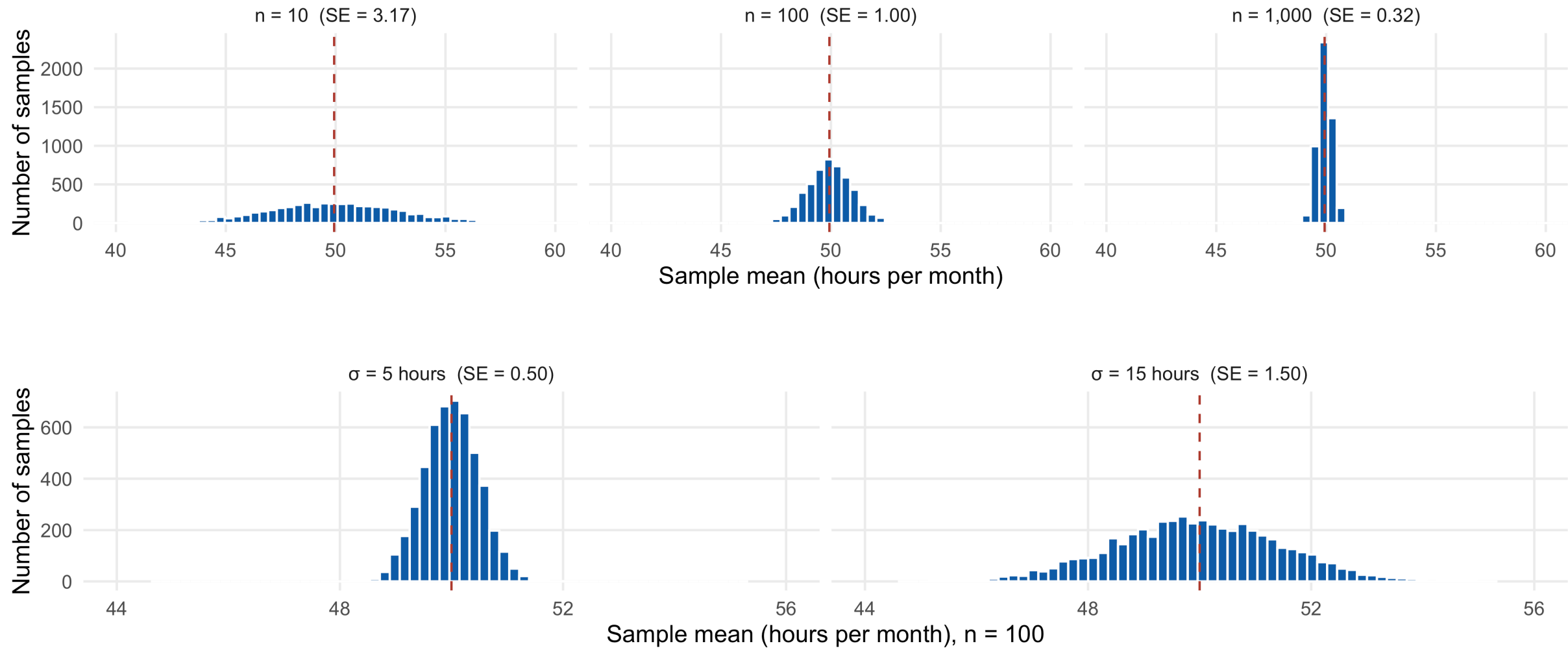
**Square-root law:** four times the sample, half the standard error — **diminishing returns**. This is why polls survey ~1,000 people, not 10,000.

**Four times the sample size, half the standard error**

Standard error of the mean for a population SD of  $\sigma = 10$  hours



# What drives the standard error



**Larger samples** → sample means cluster more tightly. **More variable populations** → sample means scatter more.

# Live demo: what drives the standard error?

Course book, §3.4 — move the sliders for  $n$  and  $\sigma$ .

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-se>

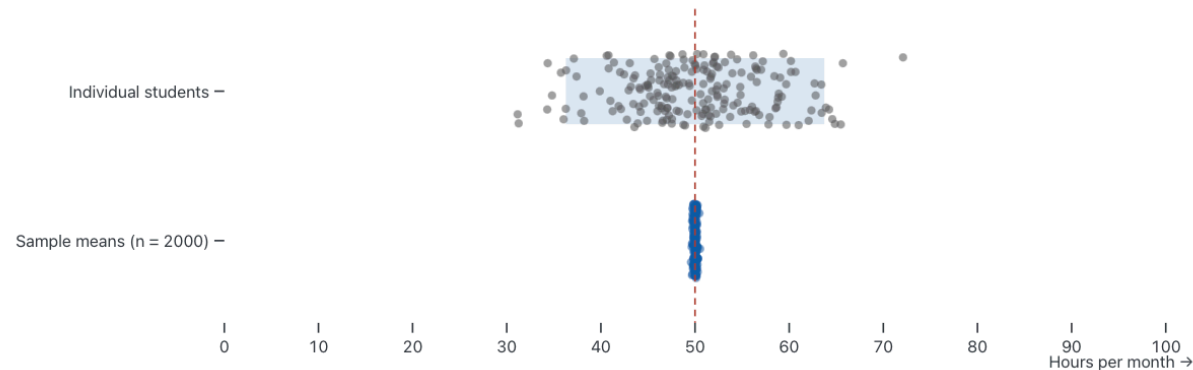
Your survey has 100 respondents and a standard error of 2 points. Your boss wants **1** point. How many respondents do you need?

## Try it: what drives the standard error?

Change the **sample size** and the **variability in the population**. The upper row shows 200 individual students; the lower row shows the means of 200 samples of size  $n$ . The bars mark the middle 95% of each distribution.

Sample size  $n$

Population SD  $\sigma$  (hours)

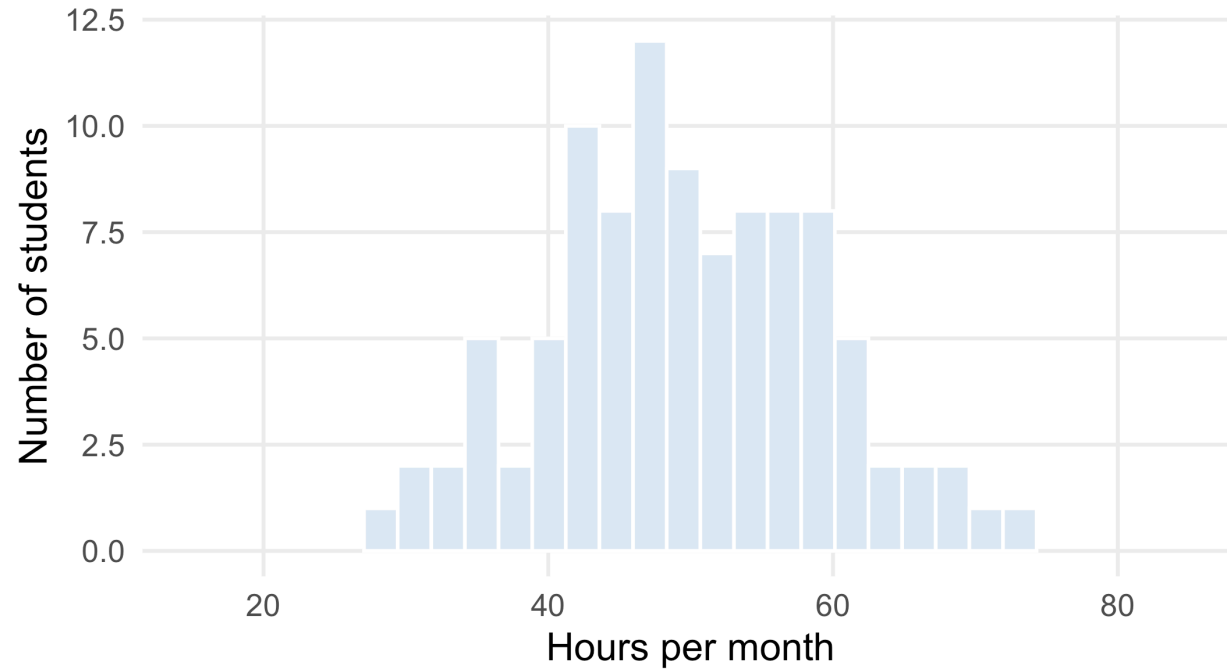


Standard deviation of individuals:  $\sigma = 7$  hours · standard error of the mean:  $\sigma / \sqrt{n} = 7 / \sqrt{2000} = 0.16$  hours · to halve the standard error, you would need  $n = 8,000$

# Standard deviation vs. standard error

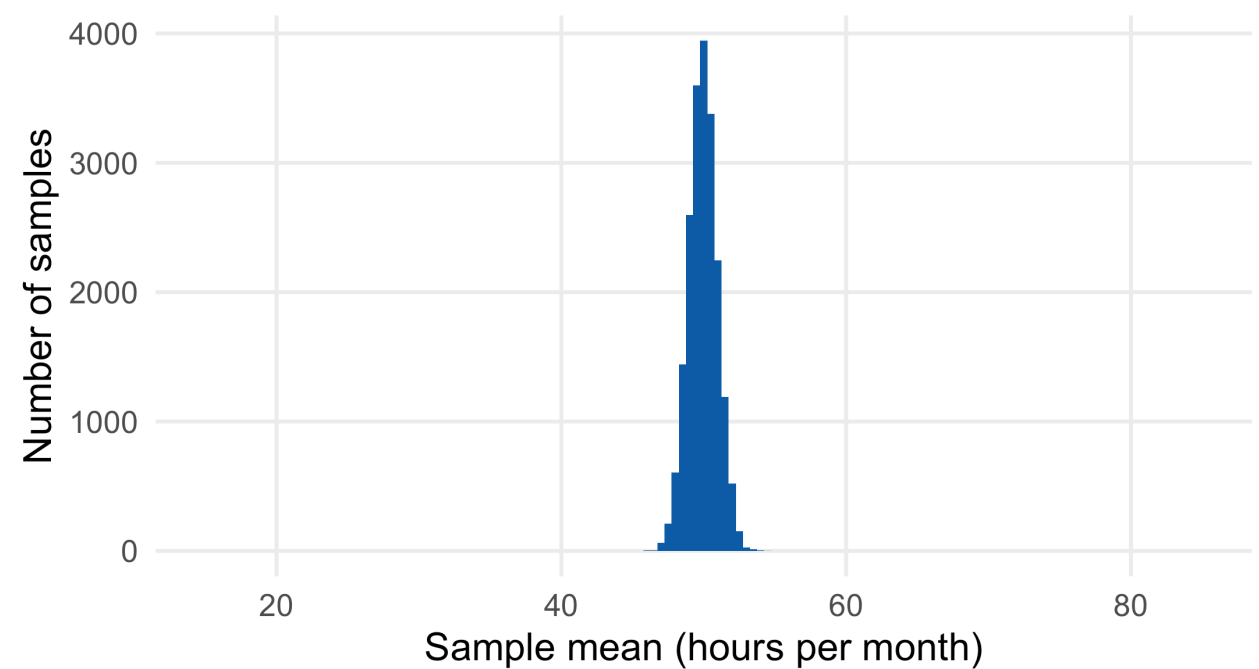
## One sample: how students differ

Standard deviation  $s = 9.4$  hours



## Many samples: how sample means differ

Standard error = 1.0 hours



**SD:** how much **individuals** differ. A property of the data — does *not* shrink with more data.

**SE:** how much **sample means** would differ. The precision of the estimate — shrinks with  $\sqrt{n}$ .

## In practice: estimate $\sigma$ with $s$

We never know  $\sigma$  — so we use the sample standard deviation  $s$ :

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Our first sample:  $s = 10.44 \rightarrow SE = 1.04$  hours (true value: 1.00).

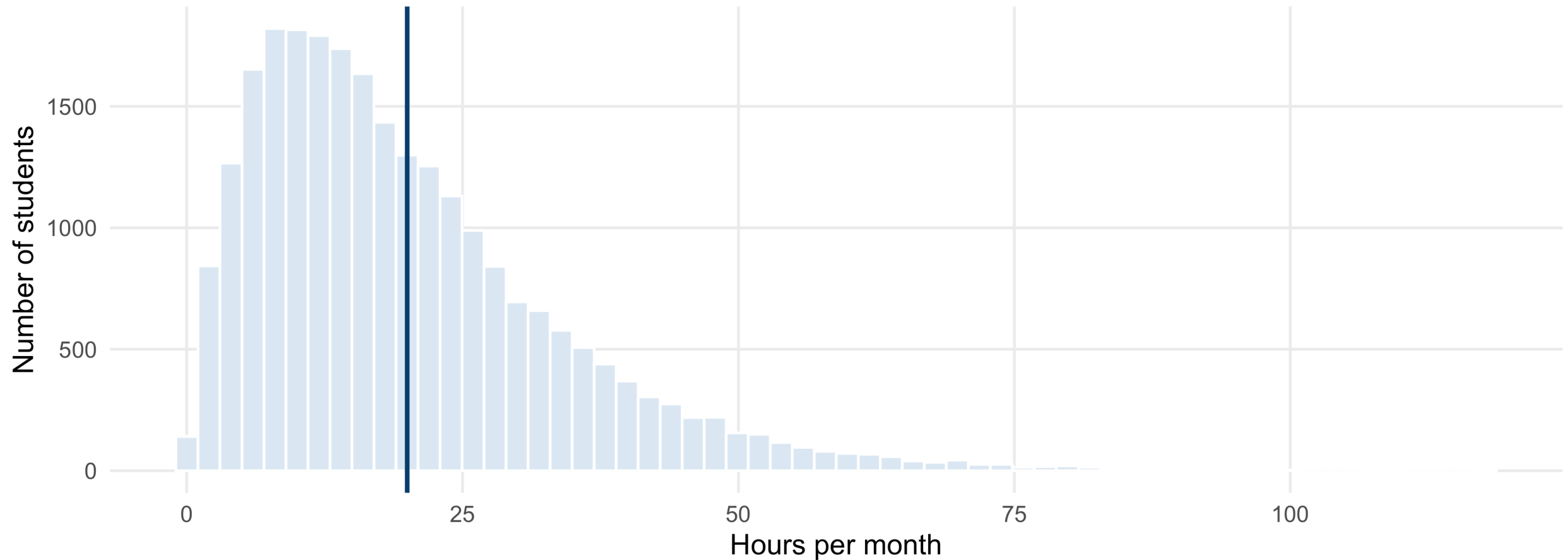
**Which one to report?** Describe how customers differ  $\rightarrow$  **SD**. Describe how precisely you estimated the mean  $\rightarrow$  **SE** or a **confidence interval**. Error bars in charts must say which one they show.

## 3.5 The central limit theorem

# What if the data are not normal?

## A right-skewed population of listening times

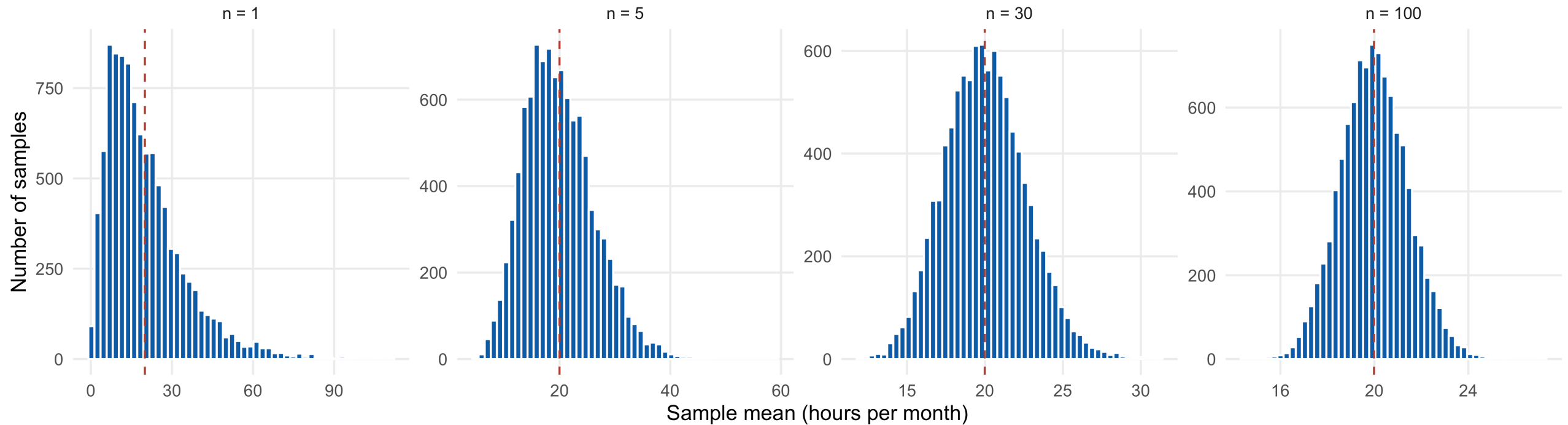
Population mean  $\mu = 19.98$  hours; population SD  $\sigma = 14.15$  hours



Marketing data are rarely normal: sales, visits, streams, revenue — **most customers a little, a few a lot.**

# Sample means become normal anyway

Distribution of 10,000 sample means for different sample sizes  $n$  (skewed population)



**Central limit theorem:** whatever the shape of the population, the **means** of samples become normally distributed as  $n$  grows — centered on  $\mu$ , with width  $\sigma/\sqrt{n}$ .

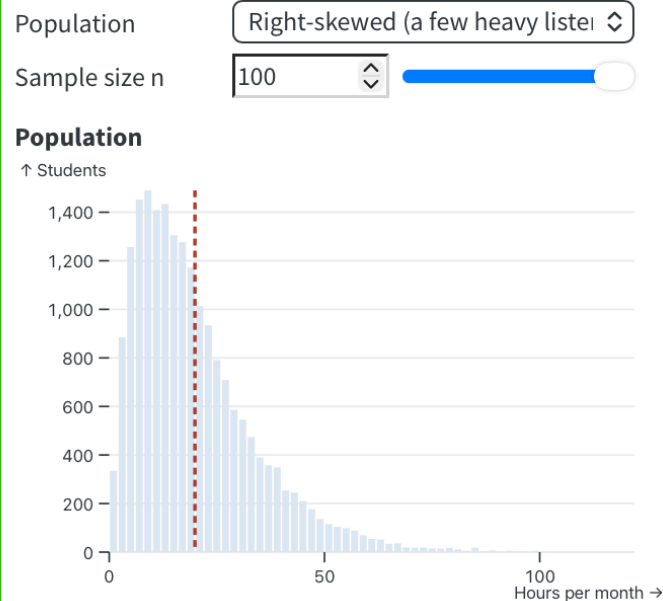
# Live demo: the central limit theorem

Course book, §3.5 — choose a population shape, increase  $n$  step by step.

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-clt>

## Try it: the central limit theorem

Choose the shape of the population and the sample size  $n$ . The left chart shows the population; the right chart shows the means of 3,000 samples of size  $n$ , together with the normal distribution predicted by the central limit theorem (orange line). Try it: the central limit theorem (orange line), how large does  $n$  have to be before the orange line fits?



**Why it matters:** we do not need normal *data* — we need a normal *sampling distribution*. “ $n \geq 30$ ” is a rule of thumb, not a law: for extremely skewed data (streams!) you need much more.

## 3.6 Confidence intervals

# From a point to an interval

**Point estimate:** “the average listening time is 49.2 hours”

**Interval estimate:** “... between 47.2 and 51.3 hours”

**Casting a net:** we cannot see the true mean — but we can throw a net around our estimate that catches it most of the time.

$$\text{CI} = \text{estimate} \pm \underbrace{\text{critical value}}_{\text{how sure we want to be}} \times \underbrace{\text{standard error}}_{\text{how much the estimate wobbles}}$$

Net metaphor after Radziwill (2015), *Statistics (The Easier Way) with R*.

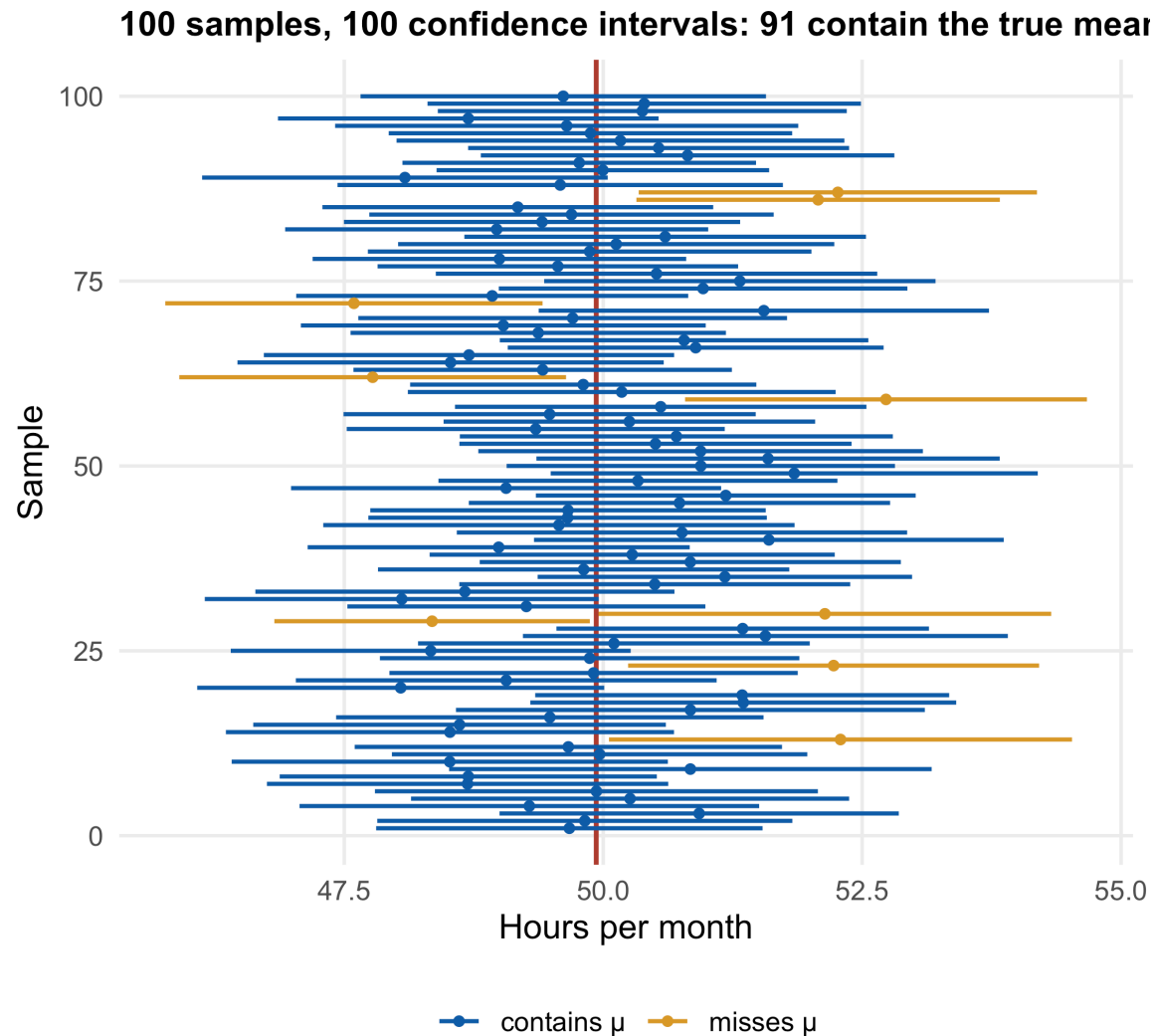
## Constructing a 95% confidence interval

- Sample means are normal around  $\mu$  with  $SD = SE \rightarrow 95\%$  lie within  $\pm 1.96 SE$
- Turned around:  $\bar{x} \pm 1.96 \times SE$  catches  $\mu$  in 95% of all samples

**Our sample:**  $49.24 \pm 1.96 \times 1.04 = [47.19; 51.29]$  hours

True mean: 49.93 — inside. This time.

# What “95% confidence” means



The confidence refers to the **method**: of all intervals constructed this way, **~95% contain** the true mean.

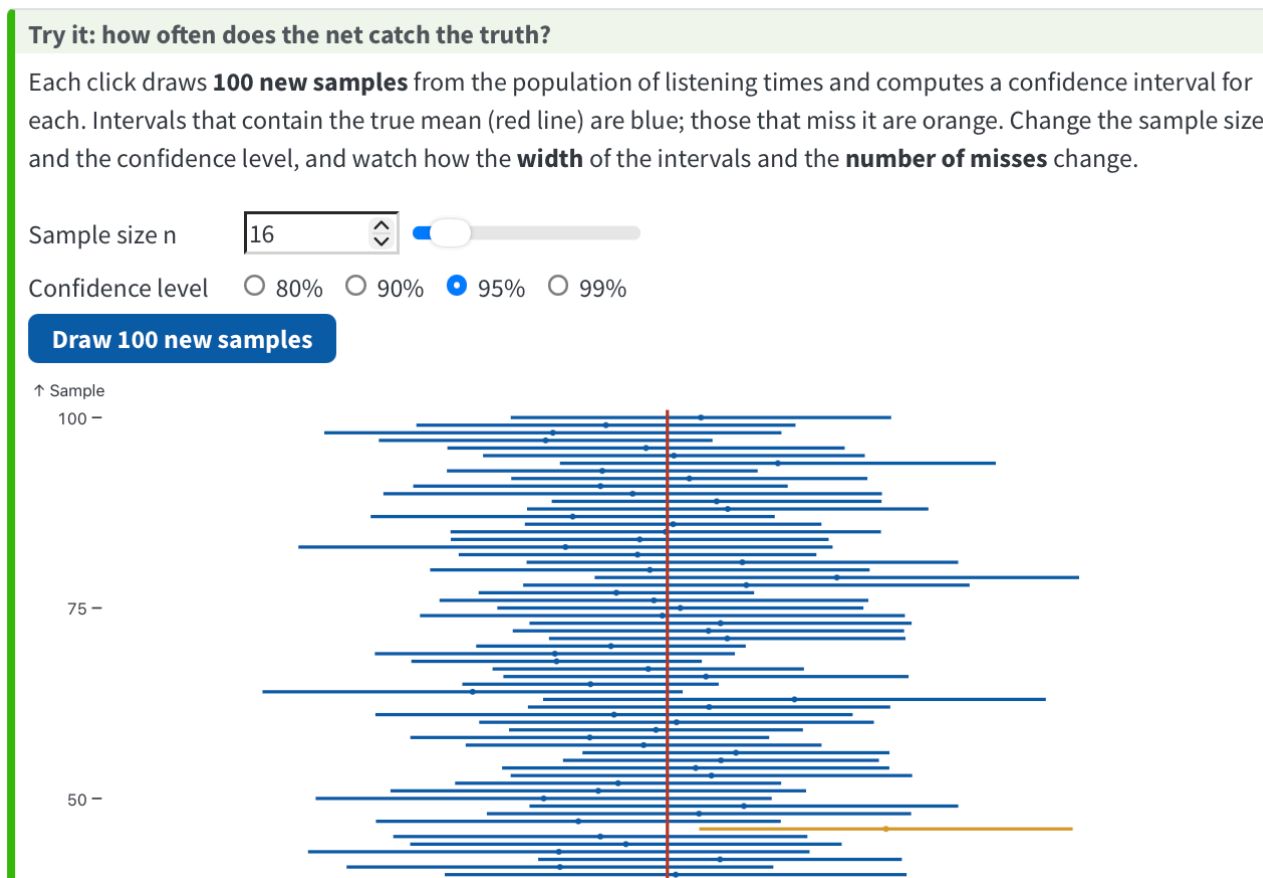
**Not quite right:** “95% probability that  $\mu$  is in *this* interval” — a specific interval either contains  $\mu$  or not.  
**Fine in practice:** “we are 95% confident that ...” — remembering that ~1 in 20 such intervals is wrong.

# Live demo: how often does the net catch the truth?

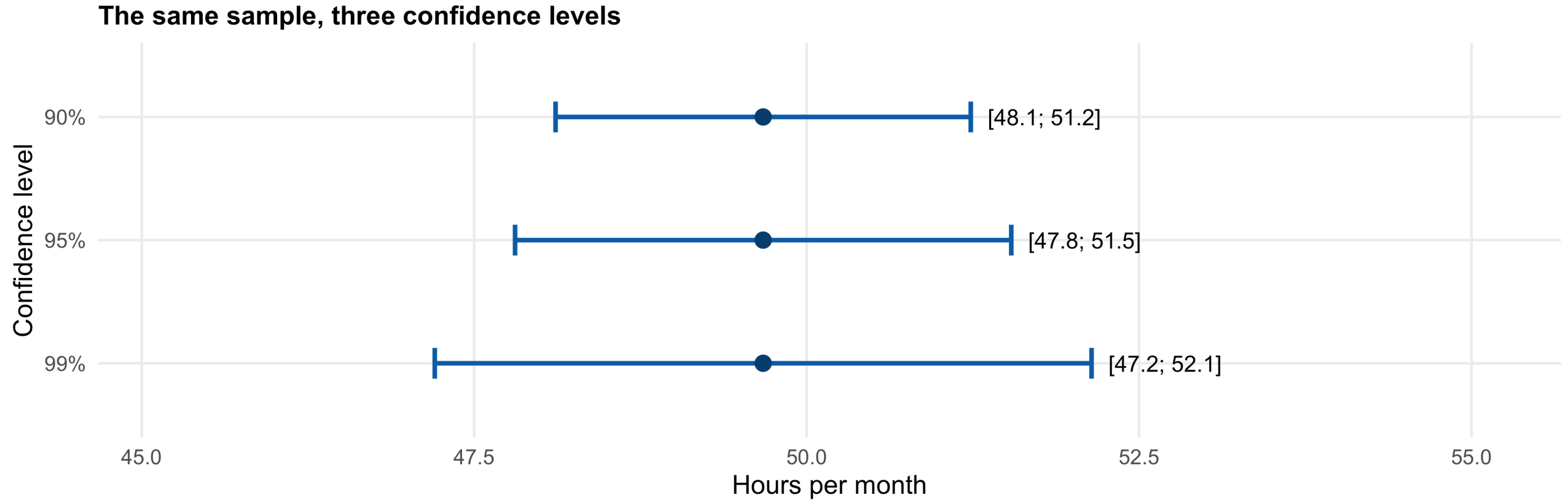
Course book, §3.6 — draw batches of 100 intervals; change  $n$  and the confidence level.

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-ci>

What happens to the **width** and to the **number of misses** when you (a) increase  $n$ , (b) switch from 95% to 99%?



# Confidence level vs. width



- More confidence → **wider** net (99%: 2.58 instead of 1.96)
- Narrow **and** reliable → only via **larger n**

| <i>n</i>             | 10   | 30   | 100  | ∞    |
|----------------------|------|------|------|------|
| <i>t</i> value (95%) | 2.26 | 2.05 | 1.98 | 1.96 |

Small samples: *t*-distribution — R does it automatically

## 3.7 Shares and the margin of error

# Decoding “± 3.1 percentage points”

Standard error of a share:

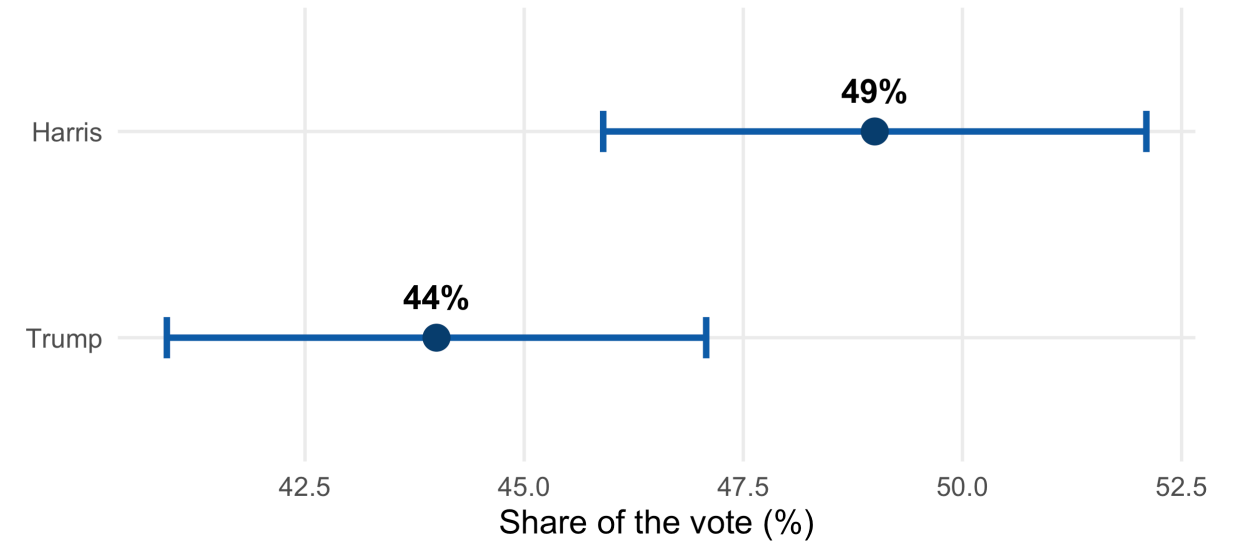
$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Margin of error =  $1.96 \times SE_{\hat{p}}$ , largest at  $\hat{p} \approx 50\%$ :

$$1.96 \times \sqrt{\frac{0.5 \times 0.5}{1,000}} = \mathbf{0.031}$$

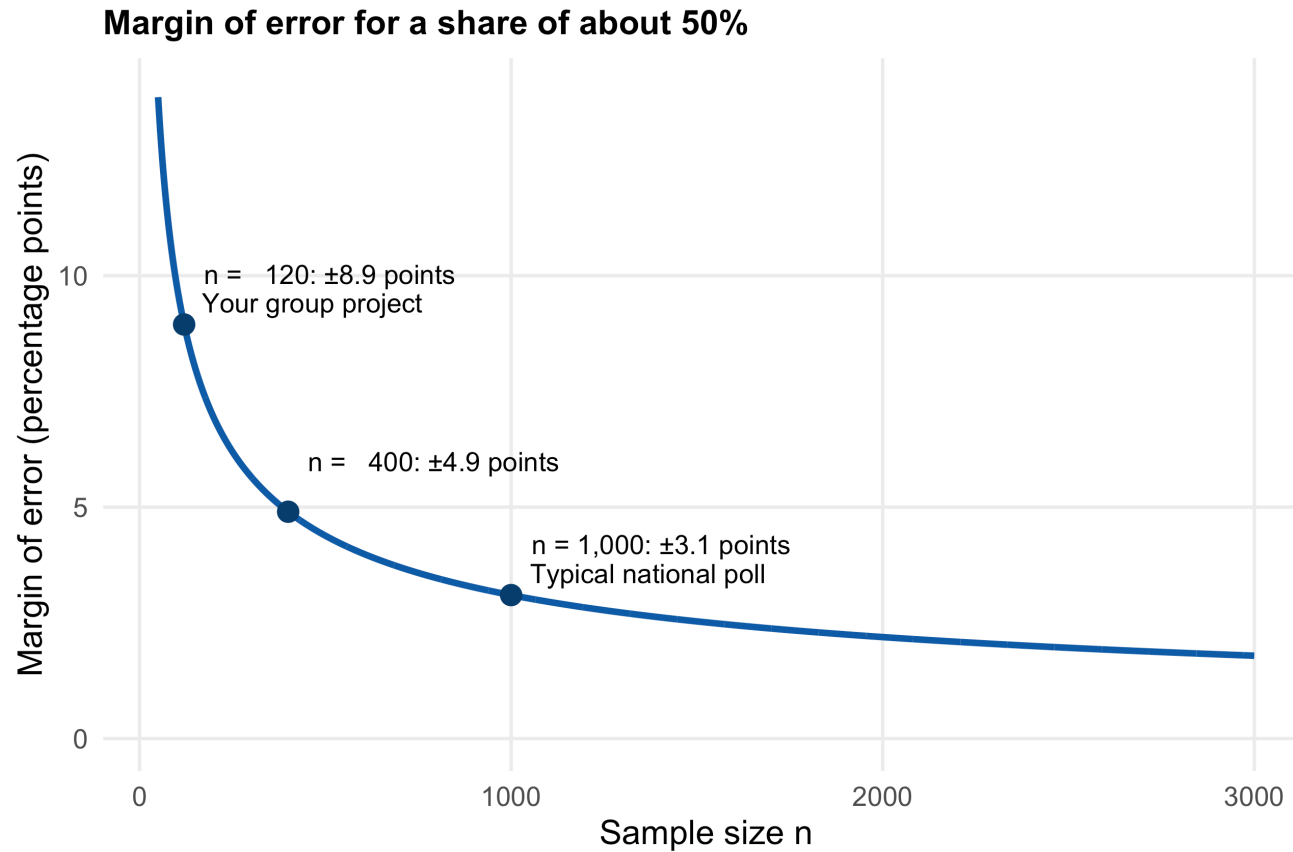
## A poll of 1,000 voters: 95% confidence intervals

The intervals overlap — but that alone does not settle whether the lead is real.



The margin of error of the **lead** is roughly **twice** as large: even the 5-point lead in this poll is not a safe lead.

# How many respondents do you need?



**Your group project** (~120 respondents): a share of ~50% comes with **± 9 points**. Think twice before interpreting a 5-point difference.

Live calculator: course book, §3.7

<https://wu-rds.github.io/MA2026/03-uncertainty.html#sec-unc-proportions>

## 3.8 Uncertainty in R

# A confidence interval for a mean

```
1 set.seed(6789)
2 population <- rgamma(25000, shape = 2, scale = 10) # the "unknown" population
3 listening <- tibble(hours = sample(population, size = 100))
```

```
1 n <- nrow(listening)
2 xbar <- mean(listening$hours)
3 se <- sd(listening$hours) / sqrt(n)
4 crit <- qt(0.975, df = n - 1)
5 c(mean = xbar, lower = xbar - crit * se, upper = xbar + crit * se)
```

```
      mean    lower    upper
19.77946 16.82950 22.72943
```

```
1 t.test(listening$hours)$conf.int
```

```
[1] 16.82950 22.72943
attr(,"conf.level")
[1] 0.95
```

`qt(0.975, ...)`: 2.5% in each tail. `t.test()` does it all in one line.

# Back to the A/B test: intervals per group

```
1 ab_test |>
2   group_by(group) |>
3   summarise(n = n(), mean = mean(hours), se = sd(hours) / sqrt(n),
4             lower = mean - qt(0.975, n - 1) * se, upper = mean + qt(0.975, n - 1) * se)
```

```
# A tibble: 2 × 6
  group      n mean   se lower upper
<fct>   <int> <dbl> <dbl> <dbl> <dbl>
1 Smart Mix    200  23.2  1.15  21.0  25.5
2 Control     200  20.0  0.968  18.1  21.9
```

The two intervals **overlap** — no effect? Not necessarily: the decision depends on the uncertainty of the **difference**, which has its own standard error. The interval `t.test()` reported (0.29 to 6.20 hours) is exactly that → **Experiments I**.

# A confidence interval for a share

Share of students listening > 30 hours:

```
1 p_hat <- mean(listening$hours > 30)
2 se_p  <- sqrt(p_hat * (1 - p_hat) / n)
3 c(share = p_hat,
4   lower = p_hat - 1.96 * se_p,
5   upper = p_hat + 1.96 * se_p)
```

| share     | lower     | upper     |
|-----------|-----------|-----------|
| 0.2200000 | 0.1388077 | 0.3011923 |

The same with `prop.test()`:

```
1 prop.test(x = sum(listening$hours > 30), n = n)$conf.int
```

```
[1] 0.1458266 0.3161314
attr(,"conf.level")
[1] 0.95
```

`prop.test()` takes the number of “successes” and the sample size. Its interval is slightly wider — a small correction for whole-number counts. **Sample size planning** follows in Chapter 4.

# What confidence intervals do not tell you

## Three limits

1. **Random error only.** A biased sample gives a precise but wrong estimate — a larger biased sample, a more precise and equally wrong one (2016 polls; \$1.6; \$2.5)
2. **Precision, not spread.** A narrow interval for the *average* does not mean customers are alike
3. **A statement about a procedure.** ~1 in 20 intervals misses — compute many, and some will

**Next:** back to Smart Mix — **+3.2 hours:** larger than sampling variation alone would produce? The difference has its own standard error — **the logic of today carries over one to one**, and we read the complete *t*-test output.

# Learning check

1. Smart Mix users listened about 3 hours more. Why does this difference alone not show that Smart Mix increases listening among **all** users?
2. A poll of 1,000 voters reports  $\pm 3.1$  points. How is this number calculated? Name one error it does **not** cover.
3. You double your survey from 400 to 800 respondents. By what factor does the standard error change?
4. Random sample of 180 users: mean listening time 7.34 hours/week, SD 6.87. Compute the 95% CI (critical value 1.96).
5. Your website survey (anyone can answer) has 20,000 responses and a very narrow CI. Why might it still mislead?